

RAP Pre-Hackathon Screening — Round 1

Constrained Object Detection & Reasoning API · Document layout detection

LIVE DEMO	Runs in the browser, no setup — upload a page, get real detections and ask a question https://huggingface.co/spaces/RISHIVEL/RAP_DocLayout_DetectionD
SOURCE CODE	Training code, API, reasoning layer, Dockerfile, build log https://github.com/RISHIVELS/doc_layout_detection

WHAT WAS BUILT

Part A - detection	RT-DETR-L fine-tuned on DocLayNet. All 11 classes are non-COCO, so no pretrained class carries any of the score.
Part B - reasoning	POST /ask routes a question to the detector or refuses it. "What is the invoice total?" is refused by design - a layout detector reads structure, never text.
The refusal rule	insufficient_information is set in deterministic Python before the LLM writes anything. The model is told the verdict; it is never asked for one.
Constraints met	No agentic frameworks. Own training code. Seeded and reproducible (seed 42, full config in reports/run_metadata.json).

TRAINED WEIGHTS

Download	/releases/download/weights-v1/best.pt
Place at	weights/best.pt · 66 MB Ultralytics checkpoint

REPORTS AND EVIDENCE

Visual evaluation report	/reports/evaluation_report.pdf
Raw metrics	/reports/metrics.json
Failure-case images	/reports/failure_examples/
Build log	/memo/BUILD_LOG.md
Written memo	MEMO.pdf - in this submission folder

RUNNING IT

API endpoints	GET /health · POST /detect · POST /ask
Local setup	README.md in the repository - venv, Docker, and curl examples
Training script	/scripts/train.py